Redes Convolucionais III

Machine Learning 2 2026.1

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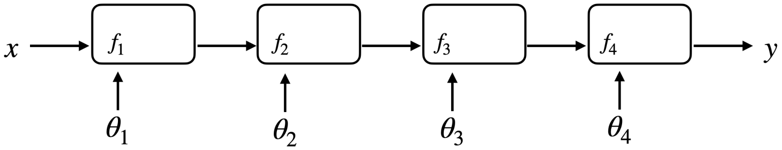
IMPA

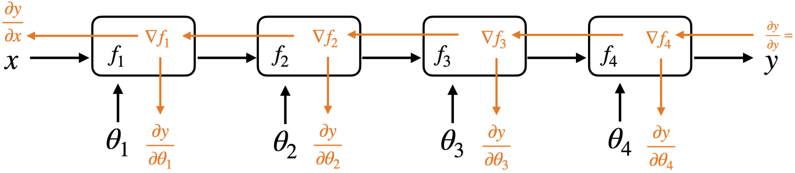
April 4, 2026

Rede Neural como Composição de Funções



$$y = f_4(f_3(f_2(f_1(x, \phi_1), \phi_2), \phi_3), \phi_4)$$

Rede Neural como Composição de Funções

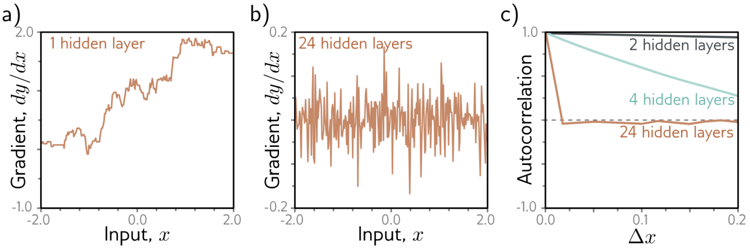


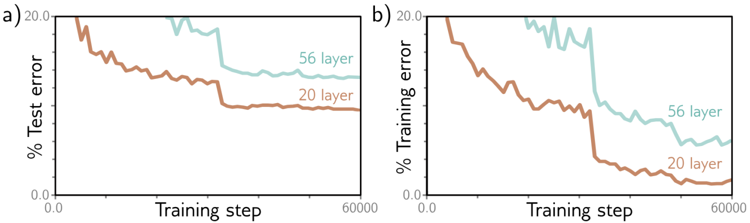
$$y = f_4(f_3(f_2(f_1(x, \phi_1), \phi_2), \phi_3), \phi_4)$$

O gradiente é dado por:

$$\frac{\partial y}{\partial \phi_1} = \frac{\partial y}{\partial f_4} \cdot \frac{\partial f_4}{\partial f_3} \cdot \frac{\partial f_3}{\partial f_2} \cdot \frac{\partial f_2}{\partial f_1} \cdot \frac{\partial f_1}{\partial \phi_1}$$

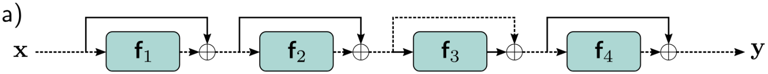
Comportamento do Gradiente





[8] 2023 — Simon Prince — *Understanding Deep Learning*

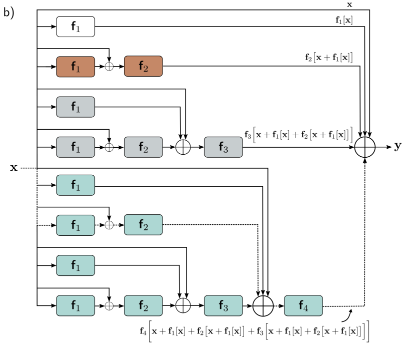
Conexões Residuais



$$
\begin{aligned}
h_1 &= x + f_1(x, \phi_1)\\
h_2 &= h_1 + f_2(h_1, \phi_2)\\
h_3 &= h_2 + f_3(h_2, \phi_3)\\
y &= h_3 + f_4(h_3, \phi_4)
\end{aligned}
$$

[8] 2023 — Simon Prince — *Understanding Deep Learning*

Conexões Residuais



$$
\begin{aligned}
y = x &+ f_1(x) \\
&+ f_2(x + f_1(x)) \\
&+ f_3(x + f_1(x) + f_2(x + f_1(x))) \\
&+ f_4(x + f_1(x) + f_2(x + f_1(x)) + f_3(x + f_1(x) + f_2(x + f_1(x))))
\end{aligned}
$$

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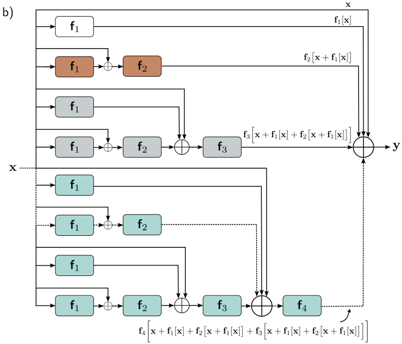
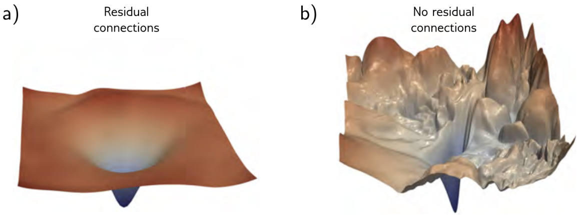
Conexões Residuais



$$\frac{\partial \mathbf{y}}{\partial \mathbf{f}_1} = \mathbf{I} + \frac{\partial \mathbf{f}_2}{\partial \mathbf{f}_1} + \left(\frac{\partial \mathbf{f}_3}{\partial \mathbf{f}_1} + \frac{\partial \mathbf{f}_3}{\partial \mathbf{f}_2}\frac{\partial \mathbf{f}_2}{\partial \mathbf{f}_1} \right) + \left(\frac{\partial \mathbf{f}_4}{\partial \mathbf{f}_1} + \frac{\partial \mathbf{f}_4}{\partial \mathbf{f}_2}\frac{\partial \mathbf{f}_2}{\partial \mathbf{f}_1} + \frac{\partial \mathbf{f}_4}{\partial \mathbf{f}_3}\frac{\partial \mathbf{f}_3}{\partial \mathbf{f}_1} + \frac{\partial \mathbf{f}_4}{\partial \mathbf{f}_3}\frac{\partial \mathbf{f}_3}{\partial \mathbf{f}_2}\frac{\partial \mathbf{f}_2}{\partial \mathbf{f}_1} \right)$$

[8] 2023 — Simon Prince — *Understanding Deep Learning*

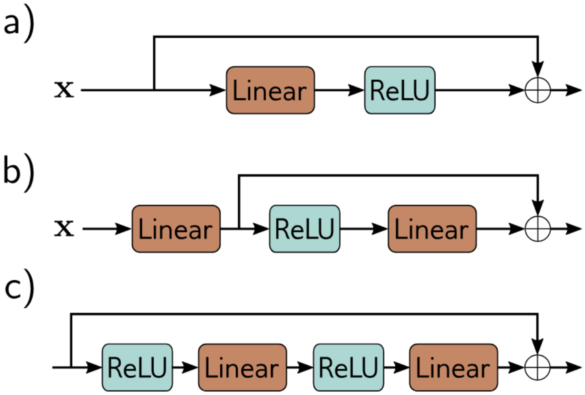
Conexões Residuais



Resnet com 56 camadas.

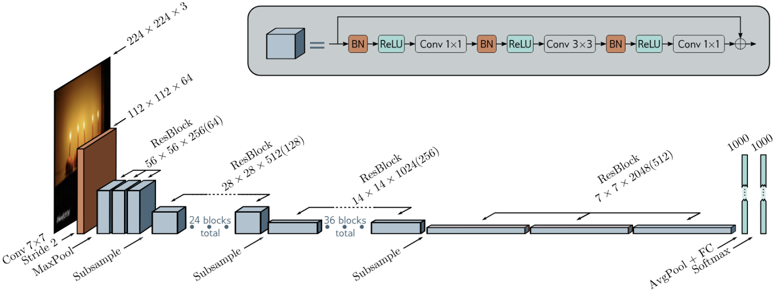
[8] 2023 — Simon Prince — *Understanding Deep Learning*

Ordem das operações



[8] 2023 — Simon Prince — *Understanding Deep Learning*

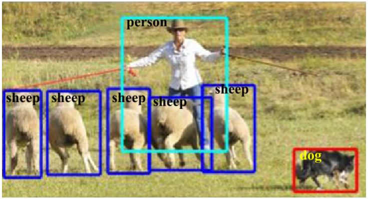
Resnet



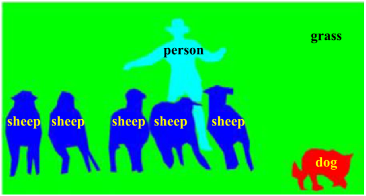
[8] 2023 — Simon Prince — *Understanding Deep Learning*
[4] 2015 — Kaiming He et al. — *Deep Residual Learning for Image Recognition*

Detecção de Objetos

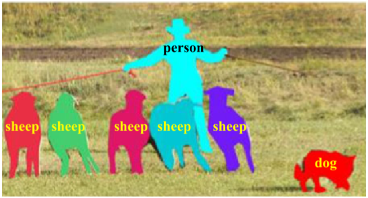
Classificação



Detecção



Segmentação

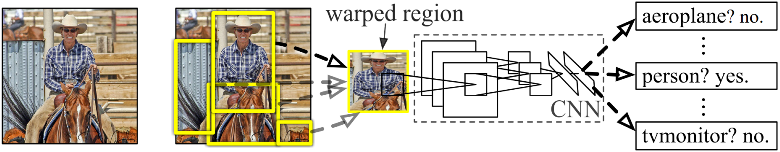


Segmentação de instâncias

[6] 2020 — Li Liu et al. — *Deep Learning for Generic Object Detection: A Survey*

Detecção de Objetos: Abordagens

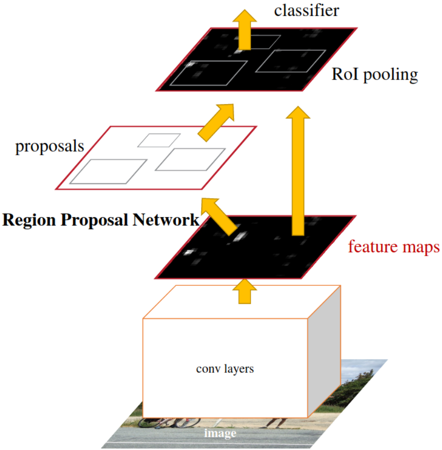
- R-CNN



[3] 2014 — Ross Girshick et al. — *Rich Feature Hierarchies for Accurate Object Detection and Semantic Segmentation*

Detecção de Objetos: Abordagens

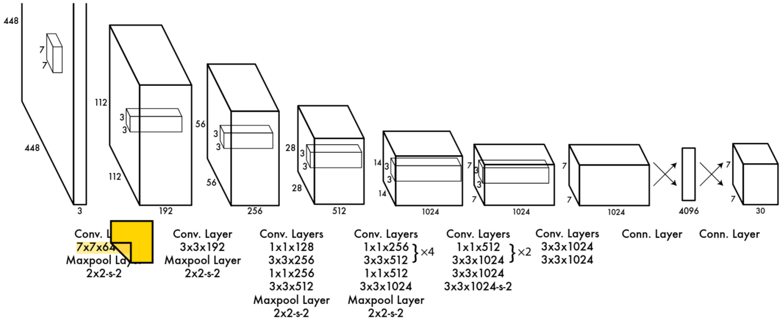
- Fast R-CNN e Faster R-CNN



[3] 2014 — Ross Girshick et al. — *Rich Feature Hierarchies for Accurate Object Detection and Semantic Segmentation*
[2] 2015 — Ross Girshick — *Fast R-CNN*
[10] 2015 — Shaoqing Ren et al. — *Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks*

Detecção de Objetos: Abordagens

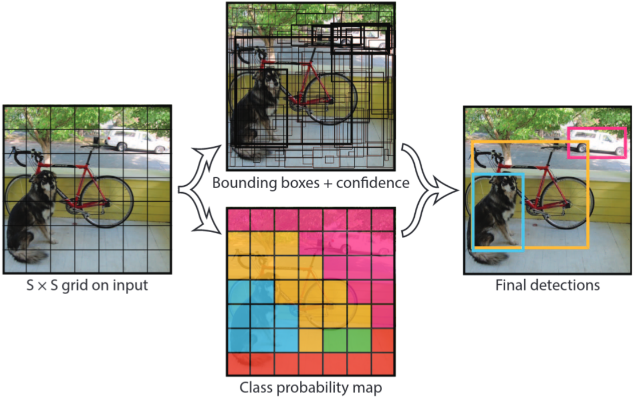
- YOLO



[3] 2014 — Ross Girshick et al. — *Rich Feature Hierarchies for Accurate Object Detection and Semantic Segmentation*
[2] 2015 — Ross Girshick — *Fast R-CNN*
[10] 2015 — Shaoqing Ren et al. — *Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks*
[9] 2016 — Joseph Redmon et al. — *You Only Look Once: Unified, Real-Time Object Detection*

YOLO

- Saída: Tensor de tamanho $(7, 7, 30)$
- Cada célula 7×7 prevê 2 caixas delimitadoras e 20 classes
- Cada caixa delimitadora tem 5 valores: x, y, largura, altura e confiança

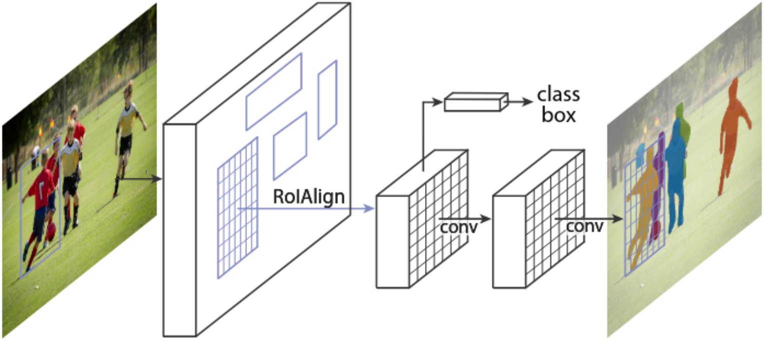


YOLO: Função de Perda

$$
\begin{aligned}
\ell = {} & \lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{\text{obj}} \left[(x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2 \right] \\
& + \lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{\text{obj}} \left[(\sqrt{w_i} - \sqrt{\hat{w}_i})^2 + (\sqrt{h_i} - \sqrt{\hat{h}_i})^2 \right] \\
& + \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{\text{obj}} (C_i - \hat{C}_i)^2 \\
& + \lambda_{\text{noobj}} \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{\text{noobj}} (C_i - \hat{C}_i)^2 \\
& + \sum_{i=0}^{S^2} \mathbb{1}_{i}^{\text{obj}} \sum_{c \in \text{classes}} \big(p_i(c) - \hat{p}_i(c) \big)^2
\end{aligned}
$$

Segmentação: Arquiteturas

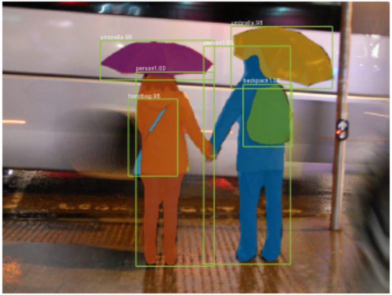
- Mask R-CNN



[5] 2018 — Kaiming He et al. — *Mask R-CNN*

Segmentação: Arquiteturas

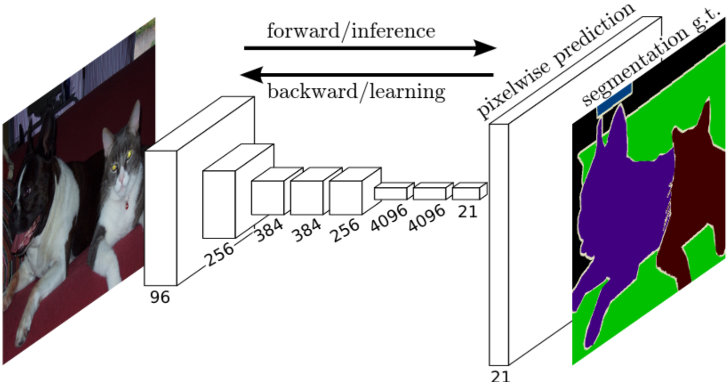
- Mask R-CNN



[5] 2018 — Kaiming He et al. — *Mask R-CNN*

Segmentação: Arquiteturas

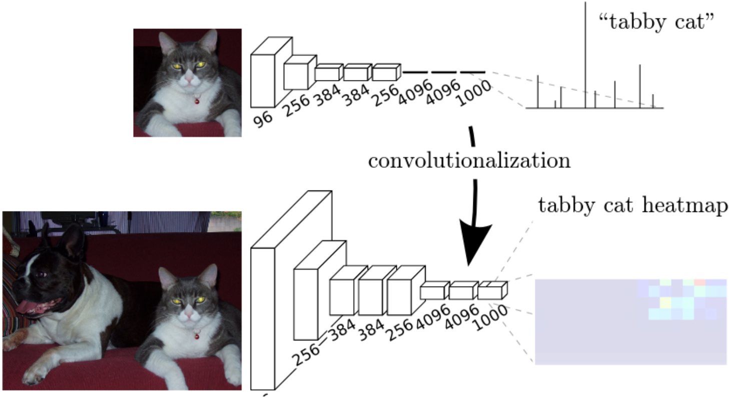
- Fully Convolutional Network (FCN)



[5] 2018 — Kaiming He et al. — *Mask R-CNN*
[7] 2015 — Jonathan Long et al. — *Fully Convolutional Networks for Semantic Segmentation*

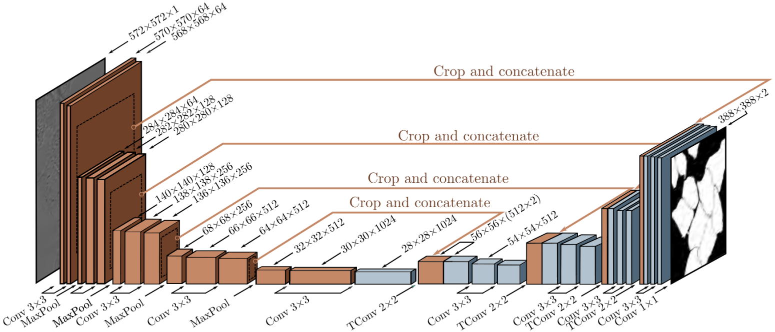
Segmentação: Arquiteturas

- Fully Convolutional Network (FCN)



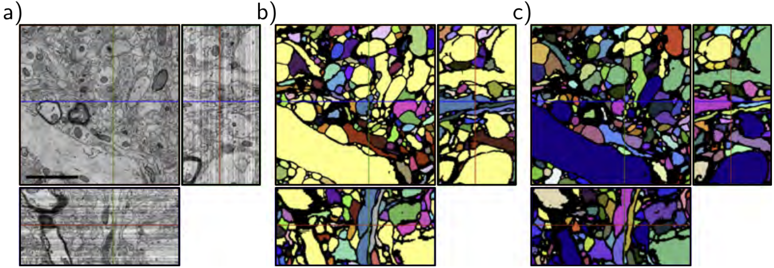
[5] 2018 — Kaiming He et al. — *Mask R-CNN*
[7] 2015 — Jonathan Long et al. — *Fully Convolutional Networks for Semantic Segmentation*

Segmentação: U-Net



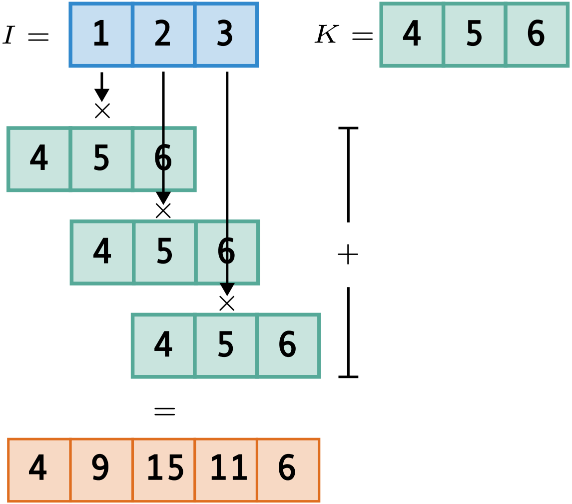
[11] 2015 — Olaf Ronneberger et al. — *U-Net: Convolutional Networks for Biomedical Image Segmentation*
[8] 2023 — Simon Prince — *Understanding Deep Learning*

Segmentação: U-Net



[11] 2015 — Olaf Ronneberger et al. — *U-Net: Convolutional Networks for Biomedical Image Segmentation*
[8] 2023 — Simon Prince — *Understanding Deep Learning*

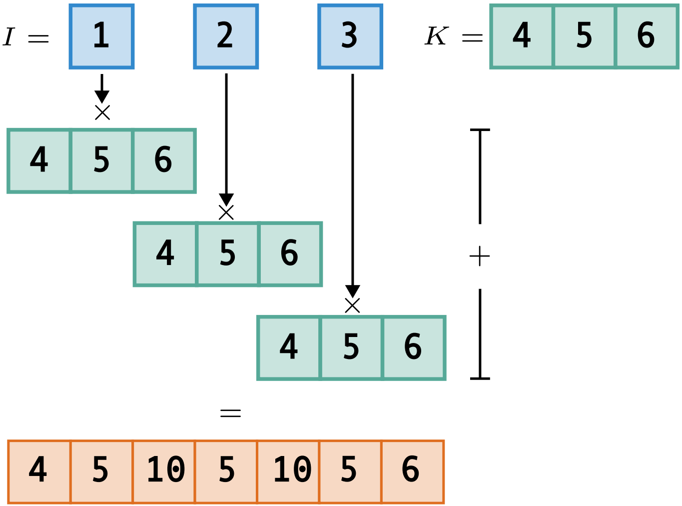
Upsampling: Convolução Transposta



[1] — *ConvTranspose2d — PyTorch 2.6 documentation*

PyTorch: [1]

Upsampling: Convolução Transposta + Stride



[1] — *ConvTranspose2d — PyTorch 2.6 documentation*

Referências I

[1] *ConvTranspose2d — PyTorch 2.6 documentation*. Read_Status: New Read_Status_Date: 2025-03-31T14:05:45.654Z. URL: https://pytorch.org/docs/stable/generated/torch.nn.ConvTranspose2d.html (visited on 03/31/2025).

[2] Ross Girshick. “Fast R-CNN”. In: *2015 IEEE International Conference on Computer Vision (ICCV)*. IEEE, Dec. 2015, pp. 1440–1448. ISBN: 978-1-4673-8391-2. DOI: 10.1109/ICCV.2015.169. URL: https://github.com/rbgirshick/ (visited on 11/24/2021).

[3] Ross Girshick et al. “Rich Feature Hierarchies for Accurate Object Detection and Semantic Segmentation”. In: *2014 IEEE Conference on Computer Vision and Pattern Recognition*. Vol. 1. IEEE, June 2014, pp. 580–587. ISBN: 978-1-4799-5118-5. DOI: 10.1109/CVPR.2014.81. arXiv: 1311.2524. URL: http://arxiv..

Referências II

[4] Kaiming He et al. *Deep Residual Learning for Image Recognition*. Dec. 10, 2015. DOI: 10.48550/arXiv.1512.03385. arXiv: 1512.03385[cs]. URL: http://arxiv.org/abs/1512.03385 (visited on 04/04/2026).

[5] Kaiming He et al. *Mask R-CNN*. Read_Status: New Read_Status_Date: 2025-04-04T19:19:09.568Z. Jan. 24, 2018. DOI: 10.48550/arXiv.1703.06870. arXiv: 1703.06870[cs]. URL: http://arxiv.org/abs/1703.06870 (visited on 04/04/2025).

[6] Li Liu et al. "Deep Learning for Generic Object Detection: A Survey". In: *International Journal of Computer Vision* 128.2 (2020), pp. 261–318. ISSN: 15731405. DOI: 10.1007/s11263-019-01247-4. arXiv: 1809.02165.

[7] Jonathan Long, Evan Shelhamer, and Trevor Darrell. *Fully Convolutional Networks for Semantic Segmentation*. Read_Status: To Read Read_Status_Date: 2025-04-04T19:25:12.957Z. Mar. 8, 2015. DOI: 10.48550/arXiv.1411.4038. arXiv: 1411.4038[cs]. URL: http://arxiv.org/abs/1411.4038 (visited on 04/04/2025).

Referências III

[8] Simon Prince. *Understanding Deep Learning*. 2023. URL: https://udlbook.github.io/udlbook/ (visited on 12/20/2023).

[9] Joseph Redmon et al. "You Only Look Once: Unified, Real-Time Object Detection". In: *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, June 2016, pp. 779–788. ISBN: 978-1-4673-8851-1. DOI: 10.1109/CVPR.2016.91. arXiv: 1506.02640. URL: http://arxiv.org/abs/1506.02640.

[10] Shaoqing Ren et al. "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks". In: *Advances in Neural Information Processing Systems* 39.6 (June 4, 2015). Ed. by C Cortes et al., pp. 159–183. ISSN: 01628828. DOI: 10.1109/TPAMI.2016.2577031. arXiv: 1506.01497. URL: http://arxiv.org/abs/1506.01497 (visited on 11/23/2021).

Referências IV

[11] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. *U-Net: Convolutional Networks for Biomedical Image Segmentation*. Read_Status: New Read_Status_Date: 2024-08-19T12:43:08.878Z. May 18, 2015. DOI: 10.48550/arXiv.1505.04597. arXiv: 1505.04597 [cs]. URL: http://arxiv.org/abs/1505.04597 (visited on 08/14/2024).